

U.G.C. NET Exam
COMPUTER
(Solved with Expl)

1. How many committees of five people can be chosen from 20 men and 12 women such that each committee contains at least three women?

- (a) 75240 (b) 52492
(c) 41800 (d) 9900

Ans : (b) For that, we have 20 men & 12 women

Case 1 : Take 2 men from 20 men & 3 women from 12 women ${}^{20}C_2 \times {}^{12}C_3$

Case 2 : Take 1 men from 20 men & 4 women from 12 women ${}^{20}C_1 \times {}^{12}C_4$

Case 3 : Take 0 men from 20 men & 5 women from 12 women ${}^{20}C_0 \times {}^{12}C_5$

All these 3 cases are mutually exclusive

$$\text{Ans} = {}^{20}C_2 \times {}^{12}C_3 + {}^{20}C_1 \times {}^{12}C_4 + {}^{20}C_0 \times {}^{12}C_5 \\ = 52492$$

2. Which of the following statement(s) is/are false?

- (1) A connected multigraph has an Euler circuit if and only if each of its vertices has even degree
(2) A connected multigraph has an Euler Path but not an Euler circuit if and only if it has exactly two vertices of odd degree
(3) A complete graph (K_n) has a Hamilton circuit whenever $n \geq 3$
(4) A cycle over six vertices (C_6) is not a bipartite graph but a complete graph over
(a) (1) only (b) (2) and (3)
(c) (3) only (d) (4) only

Ans : (d) Cycle graphs with an even number of vertices are bipartite thus C_8 also can bipartite, hence (d) is false and answer is (d)

3. Which of the following is/are not true?

- (1) The set of negative integers is countable
(2) The set of integers that are multiples of 7 is countable
(3) The set of even integers is countable
(4) The set of real numbers between 0 and 1/2 is countable

Codes :

- (a) (1) and (3) (b) (2) and (4)
(c) (2) only (d) (4) only

Ans : (d) A set is countable if it each element can be associated with a natural number.

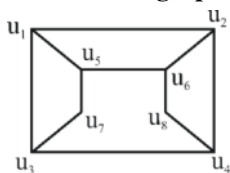
How such mapping is not possible to set of all real numbers between 0 to 1/2 hence it is not countable for rest options such mapping is possible.

n. December, 2015

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4. Consider the graph given below :



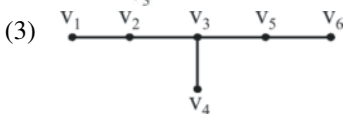
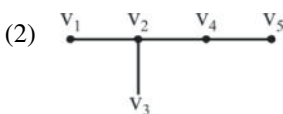
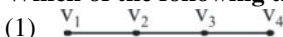
The two distinct sets of vertices, which make the graph bipartite are :

- (a) $(v_1, v_4, v_6); (v_2, v_3, v_5, v_7, v_8)$
- (b) $(v_1, v_7, v_8); (v_2, v_3, v_5, v_6)$
- (c) $(v_1, v_4, v_6, v_7); (v_2, v_3, v_5, v_8)$
- (d) $(v_1, v_4, v_6, v_7, v_8); (v_2, v_3, v_5)$

Ans : (c) A Bipartite Graph is a graph whose vertices can be divided into two independent sets, u and v such that every edge (u, v) either connects a vertex from u to v or a vertex from v to u . In other words, for every edge (u, v) , either u belongs to U and v to V , or u belongs to V and v to U .

We can say that there is no edge that connects vertices of same set $(v_1, v_4, v_6, v_7); (v_2, v_3, v_5, v_8)$ is bipartite graph vertices set.

5. A tree with n vertices is called graceful, if its vertices can be labeled with integers $1, 2, \dots, n$ such that the absolute value of the difference of the labels of adjacent vertices are all different. Which of the following trees are graceful?



Codes :

- (a) (1) and (2)
- (b) (2) and (3)
- (c) (1) and (3)
- (d) (1), (2) and (3)

Ans : (d) Caterpillar tree : In graph theory, a caterpillar is a tree in which all the vertices are within distance 1 of central path. Theorem : All caterpillar are graceful. So, (1), (2) and (3) are graceful.

6. Which of the following arguments are not valid?

- (1) "If Gora gets the job and work hard, then he will be promoted. If Gora gets promotion, then he will be happy. He will be happy, therefore, either he will not get the job or he will not work hard"

- (2) "Either Puneet is guilty or Pankaj is telling the truth. Pankaj is not telling the truth, therefore, Puneet is not guilty"
- (3) If n is a real number such that $n > 1$, then $n^2 > 1$. Suppose that $n^2 > 1$, then $n > 1$.

Codes :

- (a) (1) and (3) (b) (2) and (3)
(c) (1), (2) and (3) (d) (1) and (2)

Ans : (b)

S₁: P: Gora gets the job

Q : Gora works hard

R : Gora gets promotion

S: Gora will be happy

The argument can be written as:

$$(P \wedge Q) \rightarrow R$$

$$R \rightarrow S$$

$$\neg S$$

$$\therefore \neg P \vee \neg Q \text{ (using modus tollens)}$$

S₂: P: Puneet is not guilty

Q : Pankaj is telling the truth

The argument can be written as:

$$P \vee Q$$

$$\neg Q$$

$$\therefore P \text{ (using disjunctive syllogism) Puneet is guilty.}$$

S₃: It is not valid, as it asserts the converse of the first statement $Q \rightarrow P$, which is not equivalent to the first statement of the form $P \rightarrow Q$

Only statement (S₂) and (S₃) are not valid.

7. Let $P(m, n)$ be the statement " m divides n " where the universe of discourse for both the variables is the set of positive integers. Determine the truth values of the following propositions.

(1) $\exists m \forall n P(m, n)$

(2) $\forall n P(1, n)$

(3) $\forall m \forall n P(m, n)$

Codes :

- (a) (1) True; (2) True; (3) False
(b) (1) True; (2) False; (3) False
(c) (1) False; (2) False; (3) False
(d) (1) True; (2) True; (3) True

Ans : (a) $\forall m \forall n P(m, n)$ says that every number divides every other number & result should be a positive integer clearly (1) is false

$\forall n P(1, n)$ says that any positive integer is divisible by 1 & result will be a positive integer

2 is true

$\exists m \forall n P(m, n)$ says that there are some positive integers which divides any other positive integer

c is true

8. Match the following terms :

List - I		List - II	
(A)	Vacuous proof	(i)	A proof that the implication $p \rightarrow q$ is based on the fact that p is false

(B)	Trivial proof	(ii)	A proof that the implication $p \rightarrow q$ is true based on the fact that q is true
(C)	Direct proof	(iii)	A proof that the implication $p \rightarrow q$ is true that proceeds by showing that q must be true when p is true
(D)	Indirect proof	(iv)	A proof that the implication $p \rightarrow q$ is true that proceeds by showing that p must be false when q is false

Codes :

	(A)	(B)	(C)	(D)
(a)	(i)	(ii)	(iii)	(iv)
(b)	(ii)	(iii)	(i)	(iv)
(c)	(iii)	(ii)	(iv)	(i)
(d)	(iv)	(iii)	(ii)	(i)

Ans : (a)

List - I

(A) Vacuous proof

(i)

List - II

A proof that the implication $p \rightarrow q$ is based on the fact that p is false

(B) Trivial proof

(ii)

A proof that the implication $p \rightarrow q$ is true based on the fact that q is true

(C) Direct proof

(iii)

A proof that the implication $p \rightarrow q$ is true that proceeds by showing that q must be true when p is true

(D) Indirect proof

(iv)

A proof that the implication $p \rightarrow q$ is true that proceeds by showing that p must be false when q is false

9. Consider the compound propositions given below as :

(1) $p \wedge \sim (p \wedge q)$

(2) $(p \wedge \sim q) \vee \sim (p \wedge q)$

(3) $p \wedge (q \vee r)$

Which of the above proposition are tautologies?

- (a) (1) and (3) (b) (2) and (3)
 (c) (1) and (2) (d) (1), (2) and (3)

Ans : (b) (i) $p \vee \sim(p \wedge q) = p + (pq)'$

$$\equiv p + p' + q'$$

$$\equiv 1 + q' = 1$$

(ii) $(p \wedge \sim q) \vee \sim(p \wedge q)$

$$\equiv pq' + (pq)'$$

$$\equiv pq' + p' + q'$$

$$\equiv (p + p')(p' + q') = 1 + q'$$

$$\equiv (p' + q') + q'$$

$$\equiv p' + q' \neq 1$$

(iii) $p \wedge (q \vee r) \equiv p(q + r)$

$$\equiv pq + pr \neq 1$$

Hence (ii) and (iii) are not tautologies.

10. Which of the following property/ies a Group G must hold, in order to be an Abelian group?

- (1) The distributive property
 (2) The commutative property
 (3) The symmetric property

Codes :

- (a) (1) and (2) (b) (2) and (3)
 (c) (1) only (d) (2) only

Ans : (d) The commutative property must hold, in order to be an abelian group.

Since group is given so all property is inclusive option (d) is the answer

11. Consider the following program :

```
#include <stdio.h>
main ( )
{
    int i, inp;
    float x, term = 1, sum = 0
    scanf (" %f", & inp, & x);
    for(i = 1, <= inp; i++)
    {
        term = term * x/i;
        sum = sum + term;
    }
    printf("Result = %f\n", sum);
}
```

The program computes the sum of which of the following series?

- (a) $x + x^2/2 + x^3/3 + x^4/4 + \dots$
 (b) $x + x^2/2! + x^3/3! + x^4/4! + \dots$
 (c) $1 + x^2/2 + x^3/3 + x^4/4 + \dots$
 (d) $1 + x^2/2! + x^3/3! + x^4/4! + \dots$

Ans : (b) $i = 1$, term = 1, term = $1 * x/1$, sum = $0 + x/1$

$i = 2$, term = x , term = $x * x/2$, sum = $x + x\sqrt{2}/2$

$i = 3$, term = $x\sqrt{2}$, term = $x\sqrt{3}/6$, sum = $x + x\sqrt{2}/2 + x\sqrt{3}/6$ & do on

which is better shown $x/1! + x\sqrt{2}/2! + x\sqrt{3}/3! + \dots$

- 12. Consider the following two statements :**
- (1) A publicly derived is a subtype of base class
 - (2) Inheritance provides for code reuse
- Which one of the following statements is correct?**
- (a) Both the statements (1) and (2) are correct
 - (b) Neither of the statement (1) and (2) are correct
 - (c) Statement (1) is correct and (2) is incorrect
 - (d) Statement (1) is incorrect and (2) is correct

Ans : (a) Inheritance provide for code reuse
for public inheritance creates sub type
Inheritance only applies to user defined classes
A publicly derived class is a subtype of its base class

- 13. Consider a "CUSTOMERS" database table having a column "CITY" filled with all the names of Indian cities (in capital letters). The SQL statement that finds all cities that have "GAR" somewhere in its name, is :**
- (a) Select * from customers where city = '%GAR%';
 - (b) Select * from customers where city = '\$GAR\$'
 - (c) Select * from customers where city like '%GAR%';
 - (d) Select * from customers where city as '%GAR'

Ans : (c) Select * from customers where city like '%GAR%';

- 14. Match the following database terms to their functions :**

List - I		List – II	
(A)	Normalization	(i)	Enforces match of primary key to foreign key
(B)	Data Dictionary	(ii)	Reduce data redundancy in a database
(C)	Referential Integrity	(iii)	Defines view (s) of the database for particular user (s)
(D)	External Schema	(iv)	Contains metadata describing database structure

Codes :

- | | | | |
|----------|-------|-------|-------|
| (A) | (B) | (C) | (D) |
| (a) (iv) | (iii) | (i) | (ii) |
| (b) (ii) | (iv) | (i) | (iii) |
| (c) (ii) | (iv) | (iii) | (i) |
| (d) (iv) | (iii) | (ii) | (i) |

Ans : (b)

- (a) Normalization used to reduce data redundancy in a database (ii)
- (b) Data dictionary contains the metadata for describing data structure (iv)
- (c) Referential integrity is enforced to match the primary key to foreign key (i)
- (d) External schema defines the view of the database for particular user (iii)

15. In general, in a recursive and non-recursive implementation of problem (program) :

- (a) Both time and space complexities are better in recursive than in non-recursive program
- (b) Both time and space complexities are better in non-recursive than in recursive program
- (c) Time complexity is better in recursive version but space complexity is better in non-recursive version of the program
- (d) Space complexity is better in recursive version but time complexity is better in non-recursive version of the program

Ans : (b) Recursion may not offer any better space or time complexity it only offers compact & simple to write & understand than non recursion implementation.

16. A three dimensional array in 'C' is declared as $\text{int A}[x][y][z]$. Here, the address of an item at the location $\text{A}[p][q][r]$ can be computed as follows (where w is the word length of an integer) :

- (a) $\& \text{A}[0][0][0] + w(z \times q + z \times p + r)$
- (b) $\& \text{A}[0][0][0] + w(y \times z + z \times q + r)$
- (c) $\& \text{A}[0][0][0] + w(x \times y \times p + z \times q + r)$
- (d) $\& \text{A}[0][0][0] + w(x \times y \times q + z \times p + r)$

Ans : (b) To calculate address of $\text{A}[p][q][r]$, we have to add base address of array. Then we have to pass row elements.

$$\& \text{A}[p][q][r] = \& \text{A}[0][0][0] + w(y \times z \times p + z \times q + r)$$

17. In C++, which system – provided function is called when no handler is provided to deal with an exception?

- (a) terminate ()
- (b) unexpected ()
- (c) abort ()
- (d) kill ()

Ans : (a) The function is automatically called when catch handler can be found for a thrown exception, or for some other exceptional circumstances that makes impossible to continue the exception handling process i.e. terminate ()

18. Which of the following provides the best description of an entity type?

- (a) A specific concrete object with a defined set of processes (e.g. Jatin with diabetes)
- (b) A value given to a particular attribute (e.g. height – 230 cm)

- (c) A thing that we wish to collect data about zero or more, possibly real world examples of it may exist
- (d) A template for a group of things with the same set of characteristics that may exist in the real world

Ans : (d) It is collection of entity having common attribute.

So student is an entity type which contains entities having attribute id, name & age.

Also each entity type in a database is described by a name & a list of attribute. So we may say a table is an entity type.

19. Data which improves the performance and accessibility of the database are called :

- (a) Indexes
- (b) User data
- (c) Application metadata
- (d) Data dictionary

Ans : (a) Indexes are special lookup tables that the database search engine can use to speed up data retrieval. Simply put, an index is a pointer to data in a table. An indexed in a database is very similar to an index in the back of the book.

20. A relation $R = \{A, B, C, D, E, F, G\}$ is given with following set of functional dependencies :

$F = \{AD \rightarrow E, BE \rightarrow F, B \rightarrow C, AF \rightarrow G\}$

Which of the following is a candidate key?

- (a) A
- (b) AB
- (c) ABC
- (d) ABD

Ans : (d)

Closer of given keys:

- (a) $(A)^+ = A$
- (b) $(AB)^+ = ABC$
- (c) $(ABC)^+ = ABC$
- (d) $(ABD)^+ = ABCDEFG$

Since, ABD maps all attributes of relation R so, ABD is candidate key.

21. Which of the following services is not provided by wireless access point in 802.11 WLAN?

- (a) Association
- (b) Disassociation
- (c) Error correction
- (d) Integration

Ans : (c) Service provided by wireless access point in 802.11 WLAN"

- (i) Integration
- (ii) Disassociation
- (iii) Association
- (iv) Authentication
- (v) Privacy
- (vi) distribution
- (vii) MSDU delivery

For error correction we have different policies, and correction does not provided by wireless access point in 802.11 WLAN.

22. Which of the following fields in IPv4 datagram is not related to fragmentation?

- (a) Type of service (b) Fragment offset
(c) Flags (d) Identification

Ans : (a) Service types are not in the field of IPv4 datagram. While flags, identification and fragmentation offset are related to IPv4 datagram.

23. Four channels are multiplexed using TDM. If each channel sends 100 bytes second and we multiplex 1 byte per channel, then the bit rate for the link is

- (a) 400 bps (b) 800 bps
(c) 1600 bps (d) 3200 bps

Ans : (d) Each frame carries 1 byte from each channel we have 4 channels.

The size of each frame, therefore, is $1 \times 4 = 4$ bytes or 32 bits

Each channel is sending 100 bytes/s & a frame carries 1 byte from each channel, the frame rate must be 100 frame per second. The bit rate is 100 & times, 32 = 3200 bps.

24. In a typical mobile phone system with hexagonal cells, it is forbidden to reuse a frequency band in adjacent cells. If 840 frequencies are available, how many can be used in a given cell?

- (a) 280 (b) 210
(c) 140 (d) 120

Ans : (a) Hexagonal calls means each call has 3 neighbors.

In order words, only 3 unique cells are needed consequently, each cell can have $840/3 = 280$ frequencies.

25. Using $p = 3$, $q = 13$, $d = 7$ and $e = 3$ in the RSA algorithm, what is the value of cipher text for a plain text 5?

- (a) 13 (b) 21
(c) 26 (d) 33

Ans : (*) Given, $p = 3$, $q = 13$, $d = 7$, $e = 3$, $m = 5$

As, $c = m^e \bmod n$

where $n = p.q = 3 \times 13 = 39$

$\Rightarrow c = (5)^3 \bmod 39$

$\Rightarrow c = 125 \bmod 39$

$\Rightarrow c = 8$

26. A virtual memory has a page size of 1K words. There are eight pages and four blocks. The associative memory page table contains the following entries :

Page	Block
0	3
2	1
5	2
7	0

Which of the following list of virtual addresses (in decimal) will not cause any page fault if referenced by the CPU?

- (a) 1024, 3072, 4096, 6144
- (b) 1234, 4012, 5000, 6200
- (c) 1020, 3012, 6120, 8100
- (d) 2021, 4050, 5112, 7100

Ans : (c) Page size is $1k = 1024$ words
address rang (page number)

0 – 1023 (0)
1024 – 2047 (1)
2048 – 3071 (2)
3072 – 4095 (3)
(4096 – 5119) (4)
5120 – 6143 (5)
6144 – 7167 (6)
7168 – 8191 (7)

By using above we can make string from given virtual address.

Page already in memory are 0, 2, 5, 7 now for no page fault to occur string must be 0, 2, 5, 7

Choice 1 will produce string 1, 3, 4, 6 so page fault will occur

Choice 2 will produce string 1, 3, 4, 6 so page fault will occur

Choice 4 will produce string 1, 3, 4, 6 so page fault will occur

Choice 3 will produce string 0, 2, 5, 7 hence no page faults, so ans is (c)

27. Suppose that the number of instructions executed between page faults is directly proportional to the number of page frames allocated to a program. If the available memory is doubled, the mean interval between page faults is also doubled. Further, consider that a normal instruction takes one microsecond, but if a page fault occurs, it takes 2001 microseconds. If a program takes 60 sec to run, during which time it gets 15,000 page faults, how long would it take to run if twice as much memory were available?

- (a) 60 sec
- (b) 30 sec
- (c) 45 sec
- (d) 10 sec

Ans : (c) Normal instruction takes 1 micro second
 10^{-6} sec instruction will page fault takes 2001 micro seconds so p.f. take 2000 micro seconds

Now given program takes 60 sec & there were 15000 p.f. so time taken by page faults = $15000 \times 2000 = 30$ sec, rest 30 second are consumed by program execution.

If memory is doubled then the mean interval between the page faults is also doubled. So when earlier 30 sec were needed by program instruction & 30 sec of p.f.

Now program execution takes 30 sec & p.f. will take 15 sec

so total time = $30 + 15 = 45$ sec

28. Consider a disk with 16384 bytes per track having a rotation time of 16 msec and average seek time of 40 msec. What is the time in msec to read a block of 1024 bytes from this disk?

- (a) 57 msec
- (b) 49 msec
- (c) 48 msec
- (d) 17 msec

Ans: (b) Transfer time = $((1024 \text{ bytes} * 16 \text{ ms}) / (16384 \text{ bytes})) = 1 \text{ msec}$

Rotational latency = $16 \text{ msec} / 2 = 8 \text{ msec}$

Total time = seek + rotational latency + transfer time
 $= 40 + 18 + 1 = 49 \text{ msec}$

- 29. A system has four processes and five allocatable resources. The current allocation and maximum needs are as follows :**

	Allocated	Maximum	Available
Process A	10211	11213	00x11
Process B	20110	22210	
Process C	11010	21310	
Process D	11110	11221	

The smallest value of x for which the above system is safe state is

- (a) 1 (b) 3
 (c) 2 (d) 0

Ans : (c) Remaining need = Maximum - Allocated

	Maximum	Allocated	Available	Remaining needs
A	1 1 2 1 3	1 0 2 1 1	0 0 x 1 1	0 1 0 0 2
B	2 2 2 1 0	2 0 1 1 0		0 2 1 0 0
C	2 1 3 1 0	1 1 0 1 0		1 0 3 0 0
D	1 1 2 2 1	1 1 1 1 0		0 0 1 1 1

If $x = 0$, then no process satisfied remaining needs.

If $x = 1$, then only process 'D' can be executed other processes will not satisfy remaining needs. If $x = 2$, then all processes will be execute as safe sequence is D, C, B, A.

- 30. In Unix, the login prompt can be changed by changing the contents of the file**

- (a) contrab (b) init
 (c) gettydefs (d) inittab

Ans : (c) The gettydefs file supplies the getty command with information about various terminal attributes, such as the default initialization & the login message. The gettydefs file contains single line entries that are separated by blank lines.

(i) There is nothing like contrab in unix.

(ii) The inittab file controls what happens whenever a Unix system is rebooted or forced to change run levels.

(iii) init (short for initialization) is the first process started during booting of the computer system.

- 31. A data cube C, has n dimensions, and each dimension has exactly p distinct values in the base cuboid. Assume that there are no concept hierarchies associated with the dimensions.**

What is the maximum number of cells possible in the data cube, C?

- (a) p^n (b) p
 (c) $(2^n - 1)p + 1$ (d) $(p + 1)^n$

Ans: (d) Maximum number of cells possible (including both base cells & aggregate cells) in the data cube, C : $(p + 1)^n$

So answer is option (d)

- Ans :** (b) $P(B/A) = P(A/B) \cdot P(B)/P(A)$
 Assume A refer to stiff neck & B refer to probability of meningitis,
 $P(A/B)$ = meningitis causes stiff neck = 50% = 0.5
 $P(B)$ = 1/50000
 $P(B/A) = 0.5 \times 1/50000 / (1/20) = 0.02\%$

- | |
|--|
| <p>Ans : (b) OLAP (Online Analytical Processing)</p> <ul style="list-style-type: none">• Characterized by relatively low volume of transactions.• Queries are after very complex & involve aggregations.• for OLAP systems a response time is an effectiveness measure• OLAP applications are widely used by a data mining technique |
|--|

- Ans : (a)** Four views regarding the design of a data warehouse:

 - **Top-down view:** Allows selection of the relevant information necessary for the data warehouse.
 - **Data source view:** Exposes the information being captured, stored and managed by operational systems.
 - **Data warehouse view:** Consists of fact tables and dimension tables.
 - **Business query view :** Sees the perspectives of data in the warehouse from the view of end-user.

Top-down view allows selection of the relevant information necessary for the data warehouse.

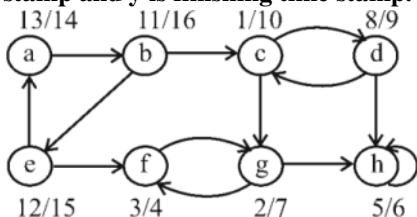
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Ans: (c) The hash function used in double hashing is of the form.

$$(h_1(k) + i \cdot h_2(k)) \bmod |T|$$

Generally, h_1 & h_2 are selected from a set of universal hash functions.

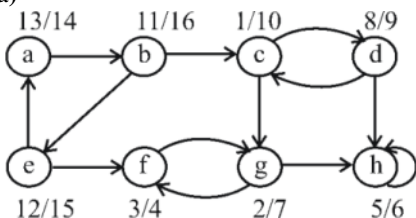
- 36. In the following graph, discovery time stamps and finishing time stamps of depth first search (DFS) are shown as x/y , x is discovery time stamp and y is finishing time stamp.**



It shows which of the following depth first forest?

- (a) $\{a, b, e\} \{c, d, f, g, h\}$
- (b) $\{a, b, e, \} \{c, d, h\} \{f, g\}$
- (c) $\{a, b, e\} \{f, g\} \{c, d\} \{h\}$
- (d) $\{a, b, c, d\} \{e, f, g\} \{h\}$

Ans : (a)



DFS visited nodes as:

$c \rightarrow g \rightarrow f \rightarrow h \rightarrow g \rightarrow d \rightarrow c$

then $b \rightarrow e \rightarrow a$

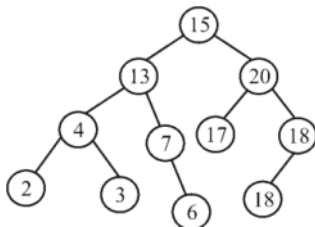
then forest is $\{a, b, e\}, \{c, d, f, g, h\}$

- 37. The number of disk pages access in B – tree search, where h is height, n is the number of keys, and t is the minimum degree is :**

- (a) $\theta(\log_n h \times t)$
- (b) $\theta(\log_t n \times h)$
- (c) $\theta(\log h n)$
- (d) $\theta(\log_t n)$

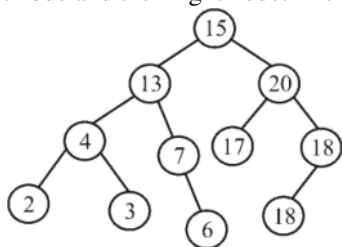
Ans : (d) The number of disk page accessed by B-TREE-SEARCH is $\theta(h) = \theta(\log + h)$, where h is height of the B-tree & h is the number of the keys in the B tree Hence option (d) is correct

- 38. The inorder traversal of the following tree is :**



- (a) 2 3 4 6 7 13 15 17 18 18 20
- (b) 20 18 18 17 15 13 7 6 4 3 2
- (c) 15 13 20 4 7 17 18 2 3 6 18
- (d) 2 4 3 13 7 6 15 17 20 18 18

Ans : (d) In inorder traversal first we traverse left node then root node and then right node: In the following tree



we first go to the left most node then its root after that right i.e. 2 4 13 7 6 15 17 20 18 18. In the rest of the option inorder property is violating.

39. An ideal sort is an in-place-sort whose additional space requirement is.....

- (a) $O(\log_2 n)$
- (b) $O(n \log_2 n)$
- (c) $O(1)$
- (d) $O(n)$

Ans : (c) In place sort are those sorts which doesn't need extra space it uses own space and in case if space is required then it is $O(1)$.

40. Which of the following is not a congestion policy at network layer?

- (a) Flow control policy
- (b) Packet discard policy
- (c) Packet lifetime management policy
- (d) Routing algorithm

Ans : (a) Flow control is function of data link layer and transport layer but not by network layer. Flow control is the process of managing the rate of data transmission between two nodes to prevent a fast sender from overwhelming a slow receiver. Routing Algorithm, Packet Lifetime Management Policy and Packet - Discard. Policy are congestion policies used in network layer. Network layer congestion policy are—

- (1) Virtual circuits vs datagram inside the sub net
- (2) Packet sequencing & service policy
- (3) Packet discard policy
- (4) Routing algorithm
- (5) Packet lifetime management.

41. Loop unrolling is a code optimization technique:

- (a) that avoids tests at every iteration of the loop
- (b) that improves performance by decreasing the number of instructions in a basic block
- (c) that exchanges inner loops with outer loops
- (d) that reorders operations to allow multiple computations to happen in parallel

Ans : (a) Loop unrolling is used to reduce the number of jump & branch instructions which could potentially make the loop faster but will increase the size of the binary. Depending on the implementation & platform, either could be faster.

42. What will be the hexadecimal value in the register ax (32-bit) after executing the following instructions?

```

mov al, 15      mov ah, 15
mov cl, 3       mov cl, 3
shr ax, cl
  
```

Codes :

- (a) 0F00 h (b) 0F0F h
(c) 01E0 h (d) FFFF h

Ans : (c) In assembly $AX = AH + BL$

AL is the lower 8 bits

AH is the higher 8 bits

here AL and AH both contains 15 or F as FX contains of or 0000 1111 0000 1111

XOR AL AL will return D in AL so AX becomes 0000 1111 0000 0000

MOV CL, 3 will store 3 in CL

Shr ax, CL will shift right ax by 3 so 0000 1111 0000 0000 will become 0000 0001 1110 0000 or 01E0 in hexadecimal i.e. C01E0H

43. Which of the following statements is false?

- (a) Top-down parsers are LL parsers where first L stands for left - to - right scan and second L stands for a leftmost derivation
(b) $(000)^*$ is regular expression that matches only strings containing an odd number of zeroes, including the empty string
(c) Bottom-up parsers are in the LR family, where L stands for left - to - right scan and R stands for rightmost derivation
(d) The class of context - free languages is closed under reversal. That is, if L is any context - free language, then the language $L^R = \{w^R : w \in L\}$ is context - free

Ans : (b) $(000)^*$ is a regular expression that matches only string containing an odd number of zeroes, including the empty string also contains 6, 12, 18, 24 which even.

So option (b) is the answer

44. System calls are usually invoked by using :

- (a) A privileged instruction
(b) An indirect jump
(c) A software interrupt
(d) Polling

Ans : (c) Privileged instruction cannot be the answer as system call is done from user mode & privileged instruction cannot be done from user mode.

45. The transfers the executable image of a C++ program from hard disk to main memory.

- (a) Compiler (b) Linker
(c) Debugger (d) Loader

Ans : (d) It is the responsibility of loader to transfer executable files from hard disk to main memory.

Debugger is a computer program that is used to test and debug other-programs (target program).

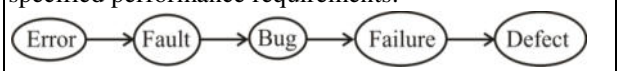
⇒ **Compiler:** Compiler is a software which converts a program written in high level language (Source Language) to low level language.

⇒ Linker takes object files, links them with each other and with any library files that you specify and produces an executable output file.

46. In software testing, how the error, fault and failure are related to each other?

- (a) Error leads to failure but fault is not related to error and failure
- (b) Fault leads to failure but error is not related to fault and failure.
- (c) Error leads to fault and fault leads to failure
- (d) Fault leads to error and error leads to failure

Ans : (c) Error is deviation from actual & expected value it represents mistake made by people. Fault is incorrect step, process or data information in a computer program which causes the program to behave in an unintended or unanticipated manner. It is the result of the error. Failure is the inability of a system or a component to a program its required functions within specified performance requirements.



```
graph LR; Error --> Fault; Fault --> Bug; Bug --> Failure; Failure --> Defect;
```

47. Which of the following is not a software process model?

- (a) Prototyping
- (b) Iterative
- (c) Timeboxing
- (d) Glassbosing

Ans : (d) Prototyping, iterative are well known software process model
In time boxing model, development is done iteratively as in iterative enhancement model each iteration is done in a timebox of fixed duration.
⇒ Glass boxing model is software testing pattern.

48. How many solutions are there for the equation $x + y + z + u = 29$ subject to the constraints that $x \geq 1, y \geq 2, z \geq 3$ and $u \geq 0$?

- (a) 4960
- (b) 2600
- (c) 23751
- (d) 8855

Ans : (b) $x + y + y + u = 29$ this is combination with repetition
 $x \geq 1, y \geq 2, z \geq 3, u \geq 0$, so allocate this to directly $29 - (1 + 2 + 3 - 0) = 29$
 $(23 + 4 - 1) C_{23} = {}^{26}C_{23} = 2600$

49. A unix file system has 1-KB blocks and 4-disk addresses. What is the maximum file size if i-nodes contain 10 direct entries and one single, double and triple indirect entry each?

- (a) 32 GB
- (b) 64 GB
- (c) 16 GB
- (d) 1 GB

Ans : (c) Disk block size = 1 KB
Disk block address = 4B
Maximum file size possible = $(DB \text{ size} / DBA)^3 \times DB \text{ size}$
 $= (1 \text{ KB}/4B)^3 \times 1 \text{ KB}$
 $= 16 \text{ GB}$

50. uses electronic means to transfer funds directly from one account to another rather than by cheque or cash

- (a) M-Banking
- (b) E-Banking
- (c) O-Banking
- (d) C-Banking

Ans : (b) Answer is E-banking. It is also known as net banking.